Optimal Portfolio Selection and Cross-Asset Market Impact under Volterra Propagators

Yann Harold N'Guessan

DUFE – Sorbonne University & École Polytechnique, Paris

Supervisor: Eduardo Abi Jaber

2026–06

Contents

1 Introduction **3**

2 Modeling **5**
2.1 Continuous-Time Model 5
2.2 Discrete-Time Model 6
2.3 Discussion on Propagator Matrix 7

3 Solution **9**
3.1 Open-Loop Solution 9
3.2 Deterministic Signal Case 11

4 Numerical Experiments **13**
4.1 Cross-Impact Influence 13
4.2 Influence of Correlation on Liquidation 16

5 Conclusion **18**

A Appendix **20**
A.1 Standard Markowitz Objective Function Derivation 20
A.2 Paper Objective Function Derivation 21
A.3 Quadratic Optimisation — General Case 21
A.4 Markowitz Optimisation Solving 21

Abstract

The paper addresses optimal portfolio selection in presence of alpha signal, execution cost (temporary impact) and market impact (transient impact). Impacts, whenever temporary or transient, are assumed to have two dimensions: impact on the traded asset itself (self-impact) and impact of other assets on the traded asset (cross-impact).

Our paper provides an alternative way to derive a solution of such risk-revenue problem by discretizing the problem and assuming deterministic trading signal. Hence, the problem simply reduces to a quadratic optimization problem. Subsequently, we do some numerical simulations to check whether observations of the continuous problem found in [1] still hold.

1 Introduction

Optimal portfolio selection is a central problem for finance practitioners. Harry Markowitz in [12] was the first one to tackle it rigorously by providing a mathematical framework for portfolio construction. His approach consists of choosing assets by balancing expected returns against risk resulting from price uncertainty. Even seven decades later, this risk-revenue approach is still a reference and continues to influence modern portfolio selection methods.

In modern optimal portfolio selection, portfolios are constructed dynamically considering not only expected returns and risk factors but also alpha signal, execution costs and market impacts. An alpha signal is a measure or a pattern that may help to predict excess returns. This information is crucial because it strongly impacts the weights attributed to each asset. Execution cost represents the extra cost added to the execution price when orders consume more liquidity than is available at the best quotes, forcing the investor to trade at progressively worse price levels. This cost disappears immediately once trading stops, hence it is called temporary impact. Execution cost is considered in portfolio selection problem because it incorporates liquidity effects - each trade reduces returns by the cost of consuming liquidity, illiquid assets being more expensive to trade. Execution cost alone is not enough for a more realistic view of orders impacts. Past trades, especially those associated with large orders, impact market participants' future quotations and thereby future returns. This price distortion caused by market participant reactions to past trading activities is called market impact. Unlike execution cost, this distortion persists after trading stops and decays gradually over time, hence it is called transient impact. Orders impacts, whether temporary or transient have two dimensions: trading asset i affects not only the price of asset i itself (self-impact) but also the price of a possible correlated asset j (Cross-impact). Modern portfolio construction approaches incorporate all these factors in optimal weight attribution.

Academically, several papers studied the incorporation of alpha signal in portfolio construction. The foundational reference is Gârleanu and Pedersen paper [8]. In their work, they provided a framework based on the decomposition of observed prices into a predictable trading signal and a pure noise. The signal was assumed to follow an Ornstein–Uhlenbeck process that decays over time reflecting alpha fading behaviour as time goes by. They showed that in presence of trading signal, the optimal strategy smoothly converges toward a time-varying Markowitz portfolio at a rate balancing alpha capture against transaction costs. Almgren et al. in [4] studied large orders impact for portfolio selection. They argued that overall impact can be decomposed into a linear temporary impact which depends on the trading speed and a permanent impact. They calibrated their model to NYSE data and found that the cost of the temporary impact dominates the total execution cost relative to the permanent component for most institutional investors. Gatheral in [10] went deeper and showed that price impact does not persist forever — it decays after a trade. He used the propagator model for measuring market impact $D_t = \int_0^t G(t-s)u_s ds$, u_s being the trading speed and G a decaying kernel. Contrary to Gârleanu and Pedersen in [9] who suggested an exponential kernel, Gatheral's works argued that the decay is well fitted by a power law kernel. This power behaviour kernel is confirmed by recent microstructure studies. Bouchaud et al. [6], [7], Benzaquen et al. [5] and Le Coz et al. [11] provided empirical finding justifying power-law decay propagator $(t-s)^{-\beta}$. Power-law decay means impact is long-memory, trades from hours ago still affect current prices. Exponential decay underestimates that persistence of time decay.

Even if exponentially decaying propagator assumption is not realistic, it has the advantage of making optimal portfolio selection problem Markovian. By doing so, standard tools such as HJB can be applied to get closed-form-solution - Gârleanu-Pedersen in [9]. Under non-exponential ker-

nel, deriving closed-form solution becomes trickier. Indeed, the problem is no longer Markovian, invalidating standard toolbox. Many attempts have been made to provide a solution under a more general kernel function assumption. However, the success of these attempts is at a cost: relaxing some constraints. For instance, Alfonsi-Klöck-Schied [2] derived a solution to the problem using general convolution kernels but only considering discrete time and without alpha signals or risk term. To the best of our knowledge, Abi Jaber et al. work [1] is the first to solve the full continuous-time multi-asset problem under a general (nonnegative-definite Volterra) propagator kernel, alpha signal, market and execution impacts. They succeeded by reducing the first-order optimality condition to a system of stochastic Fredholm integral equations of the second kind, solved via resolvent operators.

Our work uses as basis, Abi Jaber et al. work [1] and suggests a discretizing time horizon for solving the problem. By doing so, and assuming the alpha signal to be deterministic, the problem simply reduces to a quadratic optimization problem whose closed-form solution is well known. This approach has the advantage of being easy to derive, avoiding the complex mathematical developments present in [1].

The following section lays the mathematical foundations of the problem and presents the modeling steps leading to the formulation of the objective function. The next section derives an explicit solution to the optimization problem. The final section reports the results of numerical experiments and assesses their coherence with the continuous-time solution. An appendix is included at the end and provides additional details on several results used in this paper.

2 Modeling

2.1 Continuous-Time Model

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{0\le t\le T}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions of right-continuity and completeness. Let $T > 0$ be a deterministic finite time horizon and consider a portfolio of N risky assets whose unaffected price vector $(P_t)_{0\le t\le T} = (P_t^1, \dots, P_t^N)^\top \in \mathbb{R}^N$ can be decomposed as:

$$P_t \;=\; A_t \;+\; M_t \tag{1}$$

where:

- $(A_t)_{0\le t\le T} = (A_t^1, \dots, A_t^N)^\top$ is a **predictable finite-variation process** representing the forecastable drift of prices (the alpha signal).
- $(M_t)_{0\le t\le T} = (M_t^1, \dots, M_t^N)^\top$ is a **continuous martingale** representing random market noise — price innovations that cannot be forecasted. It satisfies $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for all $s \le t$, with covariance structure $d[M^i, M^j]_t = \Sigma_{ij}\, dt$, where $\Sigma \in \mathbb{R}^{N\times N}$ is a symmetric nonnegative definite covariance matrix.

The initial portfolio vector is $X_0 \in \mathbb{R}^N$. The investor trades at speed $u_t \in \mathbb{R}^N$, so that the portfolio evolves as:

$$X_t^u \;=\; X_0 \;+\; \int_0^t u_s\, ds, \quad 0 \le t \le T \tag{2}$$

The trading speed $u_t = \frac{dX_t^u}{dt} = \dot{X}_t^u$ represents the instantaneous rate of change of the portfolio - equivalently, the number of shares traded per unit of time at t. The set of admissible strategies is:

$$\mathcal{U} \;=\; \left\{ u \text{ progressively measurable } :\; \mathbb{E}\int_0^T \|u_t\|^2\, dt \;<\; \infty \right\} \tag{3}$$

Trading activity causes two effects on execution prices.

(i) Temporary impact or execution cost. The linear temporary price impact on the effective execution price is:

$$\frac{1}{2}\,\Lambda\, u_t$$

where $\Lambda \in \mathbb{R}^{N\times N}$ is a positive definite matrix (not necessarily symmetric). Its diagonal entries Λ_{ii} measure the temporary self-impact of asset i, and its off-diagonal entries Λ_{ij} measure the temporary cross-impact on asset i caused by trading one share of asset j per unit of time.

(ii) Transient or market impact. Past trades cause a persistent price distortion that decays over time. The cumulative price distortion at time t is:

$$D_t^u \;=\; \int_0^t G(t,s)\, u_s\, ds \tag{4}$$

where $G : [0,T]^2 \to \mathbb{R}^{N\times N}$ is the **Volterra propagator matrix** of kernels satisfying $G(t,s) = 0$ for $t < s$ (causality). Its diagonal entries capture linear transient self-impact, and its off-diagonal entries $G_{ij}(t,s)$ capture the impact at time t on asset i's price caused by trading asset j at time $s < t$. The nonnegative definiteness of G is required to exclude price manipulation strategies in the sense of Huberman and Stanzl.

The effective execution price is therefore $P_t + D_t^u + \frac{1}{2}\Lambda u_t$, combining the unaffected price, the accumulated transient distortion, and the instantaneous temporary impact.

The investor seeks the admissible strategy $u^* \in \mathcal{U}$ maximising the following risk-revenue functional:

$$J(u) \;=\; \mathbb{E}\Big[-\int_0^T u_t^\top \big(P_t + D_t^u + \tfrac{1}{2}\Lambda u_t\big)\; dt \;+\; (X_T^u)^\top P_T \;-\; \frac{\gamma}{2}\int_0^T (X_t^u)^\top \Sigma\, X_t^u \; dt\Big] \tag{5}$$

where $\gamma \geq 0$ is the risk-aversion coefficient. The two components of $J(u)$ are:

- $-\int_0^T u_t^\top (P_t + D_t^u + \frac{1}{2}\Lambda u_t)\; dt + (X_T^u)^\top P_T$: total wealth net of all impact costs
- $-\frac{\gamma}{2}\int_0^T (X_t^u)^\top \Sigma X_t^u \; dt$: Markowitz risk penalty on the running portfolio

2.2 Discrete-Time Model

Under a discrete-time framework, the continuous quantities introduced in Section 2.1 are replaced by their discrete counterparts. We fix $n \in \mathbb{N}$ and partition $[0, T]$ into $n+1$ equally spaced dates:

$$t_k \;=\; k\,\Delta t, \quad k = 0, 1, \ldots, n, \qquad \Delta t \;:=\; \frac{T}{n}$$

The unaffected price vector at time t_k decomposes as in the continuous case:

$$P_k \;=\; A_k \;+\; M_k \tag{6}$$

where A_k is the predictable component and M_k is a discrete martingale with $\mathbb{E}[M_{k+1} \mid \mathcal{F}_{t_k}] = M_k$ and covariance structure $\mathbb{E}[\Delta M_k \Delta M_k^\top] = \Sigma\,\Delta t$, where $\Delta M_k := M_{k+1} - M_k$.

The investor chooses trading speeds $u_k \in \mathbb{R}^N$ for $k = 0, 1, \ldots, n-1$. No trade is executed at the terminal date $t_n = T$, so $u_n = 0$ by convention. The full decision variable is the stacked vector of active trading speeds:

$$\mathbf{u} \;:=\; \begin{pmatrix} u_0 \\ u_1 \\ \vdots \\ u_{n-1} \end{pmatrix} \;\in\; \mathbb{R}^{Nn}$$

The portfolio evolves as $X_{k+1} = X_k + u_k\,\Delta t$, giving:

$$X_k \;=\; X_0 \;+\; \Delta t \sum_{j=0}^{k-1} u_j, \quad k = 0, 1, \ldots, n \tag{7}$$

This is the direct discretisation of the continuous relation $X_t^u = X_0 + \int_0^t u_s\; ds$. In particular the terminal portfolio is:

$$X_n \;=\; X_0 \;+\; \Delta t \sum_{j=0}^{n-1} u_j$$

which depends only on $u_0, \ldots, u_{n-1}$ — the Nn decision variables in $\mathbf{u}$.

The discrete linear temporary price impact at time t_k, for $k = 0, \ldots, n-1$, is:

$$\frac{1}{2}\,\Lambda\, u_k$$

The discrete price distortion at time t_k caused by all past trades is:

$$D_k \;:=\; \Delta t \sum_{j=0}^{k-1} G(t_k,\, t_j)\, u_j \tag{8}$$

This is the Riemann sum approximation of the continuous distortion $D_t^u = \int_0^t G(t,s)\, u_s\, ds$.

In the discrete problem, we add a non-liquidation penalty. It reflects the investor's wish to penalise any remaining inventory at the terminal date T. This is captured by an additional quadratic penalty on the terminal portfolio X_n:

$$-\frac{\varrho}{2}\, X_n^\top \Pi\, X_n$$

where $\varrho > 0$ is the penalty intensity and $\Pi \in \mathbb{R}^{N\times N}$ is a symmetric nonnegative definite matrix.

The goal of the investor is to maximise expected returns while penalising portfolio variance and residual inventory given market impact and execution costs. Since P_k is stochastic (it contains the martingale component M_k), the objective is taken in expectation. Replacing each continuous integral $\int_0^T f_t\, dt$ by the Riemann sum $\Delta t \sum_{k=0}^n f_k$, the discrete objective is:

$$\bar{J}(\mathbf{u}) \;=\; \mathbb{E}\left[\underbrace{-\Delta t \sum_{k=0}^{n-1} u_k^\top \big(P_k \;+\; D_k \;+\; \tfrac{1}{2}\Lambda u_k\big) \;+\; X_n^\top P_n}_{\text{total wealth net of impact costs}} \;-\; \underbrace{\frac{\gamma\,\Delta t}{2} \sum_{k=0}^{n} X_k^\top \Sigma\, X_k}_{\text{Markowitz risk penalty}} \;-\; \underbrace{\frac{\varrho}{2}\, X_n^\top \Pi\, X_n}_{\text{terminal inventory penalty}} \right] \tag{9}$$

where $\gamma \geq 0$ is the risk-aversion coefficient. The investor seeks:

$$\mathbf{u}^* \;=\; \arg\max_{\mathbf{u}\in\mathbb{R}^{Nn}} \bar{J}(\mathbf{u})$$

Remark 1. *The discrete objective* (9) *is the exact discrete analogue of the continuous objective in Remark 2.10 of [1], which incorporates the terminal penalty* $-\frac{\varrho}{2}(X_T^u)^\top \Pi X_T^u$. *The correspondence is:* $\int_0^T \to \Delta t \sum_{k=0}^n$, $X_T^u \to X_n$, $D_t^u \to D_k$, $\frac{1}{2}\Lambda u_t \to \frac{1}{2}\Lambda u_k$, *and* $\frac{\varrho}{2}(X_T^u)^\top \Pi X_T^u \to \frac{\varrho}{2} X_n^\top \Pi X_n$. *The expectation* $\mathbb{E}[\,\cdot\,]$ *is preserved in both cases because* P_k *contains the stochastic martingale component* M_k.

2.3 Discussion on Propagator Matrix

The propagator matrix $G(t,s) \in \mathbb{R}^{N\times N}$ encodes how past trades affect current prices. Following Benzaquen et al. [5] and Mastromatteo et al. [13], we adopt the parsimonious factorised specification:

$$G(t,s) \;:=\; C\,\varphi(t-s)\,\mathbf{1}_{\{t>s\}} \tag{10}$$

where:

- $C \in \mathbb{R}^{N\times N}$ is a symmetric positive semidefinite matrix of cross-impact coefficients. Its diagonal entries C_{ii} capture self-impact and its off-diagonal entries C_{ij} capture cross-impact between assets i and j.
- $\varphi : [0,T] \to \mathbb{R}$ is a scalar decay kernel controlling how fast the price distortion from a past trade recovers over time.
- $\mathbf{1}_{\{t>s\}}$ enforces causality — only past trades affect current prices.

Under this specification, Corollary 2.18 of [1] guarantees that G is nonnegative definite — and hence price manipulation is excluded — whenever $C \succeq 0$ and φ is nonnegative, nonincreasing, and convex on $(0,T)$.

Choice of scalar kernel φ. Three standard choices are considered, each capturing a different assumption on the persistence of market impact:

- **No transient** ($\varphi = 0$): trades have no persistent price impact. This is the frictionless benchmark — $D_k = 0$ at every step.
- **Exponential** ($\varphi(t) = e^{-\rho t}$): impact decays geometrically with parameter $\rho > 0$. The discrete weight of a trade m steps ago is $e^{-\rho m \Delta t}$. This kernel makes the continuous-time problem Markovian and admits closed-form solutions (Gârleanu-Pedersen [9]), but empirical studies show it underestimates impact persistence.
- **Fractional** ($\varphi(t) = t^{-\alpha}$, $\alpha \in (0, \frac{1}{2})$): impact decays as a power law, much more slowly than exponential decay. The discrete weight of a trade m steps ago is $(m\Delta t)^{-\alpha}$. This kernel is empirically better supported (Bouchaud et al. [6], Benzaquen et al. [5]) but makes the problem non-Markovian, requiring the resolvent machinery of [1].

Kernel	**Continuous formula**	**Discrete:** $\varphi(t_k - t_j)$	**Parameters**
No transient	$\varphi(t) = 0$	0	—
Exponential	$\varphi(t) = e^{-\rho t}$	$e^{-\rho(k-j)\Delta t}$	$\rho > 0$
Fractional	$\varphi(t) = t^{-\alpha}$	$\big((k-j)\Delta t\big)^{-\alpha}$	$\alpha \in (0, \frac{1}{2})$

3 Solution

This section provides steps for deriving solution of the discrete version of the problem. We provide a derivation for a general signal and then an application for a deterministic signal.

3.1 Open-Loop Solution

Expanding Each Term. Splitting (9) by effect:

$$\bar{J}(\mathbf{u}) = \mathbb{E}\left[\underbrace{-\Delta t \sum_{k=0}^{n-1} u_k^\top P_k + X_n^\top P_n}_{\mathbf{A}\text{: alpha/price}} \underbrace{- \Delta t \sum_{k=0}^{n-1} u_k^\top D_k}_{\mathbf{B}\text{: transient}} \underbrace{- \frac{\Delta t}{2} \sum_{k=0}^{n-1} u_k^\top \Lambda u_k}_{\mathbf{C}\text{: temporary}} \underbrace{- \frac{\gamma \Delta t}{2} \sum_{k=0}^{n} X_k^\top \Sigma X_k}_{\mathbf{D_1}\text{: risk}} \underbrace{- \frac{\varrho}{2} X_n^\top \Pi X_n}_{\mathbf{D_2}\text{: terminal}}\right]$$

Term A — linear in u. $-\Delta t \sum_{k=0}^{n-1} u_k^\top P_k + X_n^\top P_n = X_0^\top P_0 + \sum_{k=0}^{n-1} X_{k+1}^\top (P_{k+1} - P_k)$. This follows from:

$$\begin{cases} X_n^\top P_n - X_0^\top P_0 \;=\; \displaystyle\sum_{k=0}^{n-1} (X_{k+1}^\top P_{k+1} - X_k^\top P_k) \\ X_{k+1}^\top P_{k+1} - X_k^\top P_k \;=\; X_{k+1}^\top (P_{k+1} - P_k) + (X_{k+1} - X_k)^\top P_k \\ X_{k+1} - X_k \;=\; u_k \, \Delta t \end{cases}$$

Taking expectations, using $P_{k+1} - P_k = \Delta A_k + \Delta M_k$ and $\mathbb{E}[X_{k+1}^\top \Delta M_k] \;=\; \mathbb{E}\Big[X_{k+1}^\top \underbrace{\mathbb{E}[\Delta M_k \mid \mathcal{F}_{t_k}]}_{=0}\Big] \;=$ 0, then substituting $X_{k+1} = X_0 + \Delta t \sum_{j=0}^{k} u_j$:

$$\mathbb{E}[\mathbf{A}] = \underbrace{X_0^\top \mathbb{E}[P_0] + X_0^\top \mathbb{E}[A_n - A_0]}_{\text{constant, independent of } \mathbf{u}} + \underbrace{\Delta t \sum_{k=0}^{n-1} \left(\sum_{j=0}^{k} u_j \right)^\top \mathbb{E}[\Delta A_k]}_{\text{linear in } \mathbf{u}} \tag{11}$$

Term B — quadratic in u. Substituting $D_k = \Delta t \sum_{j<k} G(t_k, t_j) u_j$:

$$\mathbf{B} = \underbrace{-(\Delta t)^2 \sum_{k=0}^{n-1} \sum_{j=0}^{k-1} u_k^\top G(t_k, t_j)\, u_j}_{\text{quadratic in } \mathbf{u}} \tag{12}$$

Term C — quadratic in u.

$$\mathbf{C} = \underbrace{-\frac{\Delta t}{2} \sum_{k=0}^{n-1} u_k^\top \bar{\Lambda}\, u_k}_{\text{quadratic in } \mathbf{u}}, \qquad \bar{\Lambda} = \tfrac{1}{2}(\Lambda + \Lambda^\top) \tag{13}$$

Term $\mathbf{D}_1$ — linear and quadratic in u. Substituting $X_k = X_0 + \Delta t \sum_{j<k} u_j$ and expanding:

$$\begin{aligned}\mathbf{D_1} &= -\frac{\gamma\Delta t}{2}\sum_{k=0}^{n}\left(X_0 + \Delta t\sum_{j<k}u_j\right)^\top \Sigma\left(X_0 + \Delta t\sum_{j<k}u_j\right) \\ &= \underbrace{-\frac{\gamma\Delta t}{2}(n+1)\,X_0^\top\Sigma X_0}_{\text{constant}} \;-\; \underbrace{\gamma\Delta t\sum_{k=0}^{n}X_0^\top\Sigma\,\Delta t\sum_{j<k}u_j}_{\text{linear in }\mathbf{u}} \;-\; \underbrace{\frac{\gamma(\Delta t)^3}{2}\sum_{k=0}^{n}\left(\sum_{j<k}u_j\right)^\top\Sigma\left(\sum_{j<k}u_j\right)}_{\text{quadratic in }\mathbf{u}}\end{aligned} \tag{14}$$

Swapping summation $\left(\sum_{k=0}^{n}\sum_{j=0}^{k-1} = \sum_{j=0}^{n-1}\sum_{k=j+1}^{n}\right)$ and using $(n-j)\Delta t = T - t_j$:

$$-\gamma\Delta t\sum_{k=0}^{n}X_0^\top\Sigma\,\Delta t\sum_{j<k}u_j \;=\; -\gamma\Delta t\sum_{j=0}^{n-1}(T-t_j)\,u_j^\top\Sigma X_0 \tag{15}$$

So Term D_1 contributes:

$$\mathbf{D_1}: \quad \underbrace{-\frac{\gamma\Delta t}{2}(n+1)\,X_0^\top\Sigma X_0}_{\text{constant}} \;-\; \underbrace{\gamma\Delta t\sum_{j=0}^{n-1}(T-t_j)\,u_j^\top\Sigma X_0}_{\text{linear in }\mathbf{u}} \;-\; \underbrace{\frac{\gamma(\Delta t)^3}{2}\sum_{k=0}^{n}\left(\sum_{j<k}u_j\right)^\top\Sigma\left(\sum_{j<k}u_j\right)}_{\text{quadratic in }\mathbf{u}} \tag{16}$$

Term $\mathbf{D}_2$ — linear and quadratic in u. Substituting $X_n = X_0 + \Delta t\sum_{j=0}^{n-1}u_j$:

$$\mathbf{D_2}: \quad \underbrace{-\frac{\varrho}{2}X_0^\top\Pi X_0}_{\text{constant}} \;-\; \underbrace{\varrho\Delta t\sum_{j=0}^{n-1}u_j^\top\Pi X_0}_{\text{linear in }\mathbf{u}} \;-\; \underbrace{\frac{\varrho(\Delta t)^2}{2}\left(\sum_{j=0}^{n-1}u_j\right)^\top\Pi\left(\sum_{j=0}^{n-1}u_j\right)}_{\text{quadratic in }\mathbf{u}} \tag{17}$$

Full Extensive Form.

$$\begin{aligned}\bar{J}(\mathbf{u}) = {} & \underbrace{\Delta t\sum_{k=0}^{n-1}\left(\sum_{j=0}^{k}u_j\right)^\top\mathbb{E}[\Delta A_k]}_{\mathbf{A}\text{: linear in }\mathbf{u}} \;-\; \underbrace{(\Delta t)^2\sum_{k=0}^{n-1}\sum_{j=0}^{k-1}u_k^\top G(t_k,t_j)\,u_j}_{\mathbf{B}\text{: quadratic in }\mathbf{u}} \;-\; \underbrace{\frac{\Delta t}{2}\sum_{k=0}^{n-1}u_k^\top\bar{\Lambda}\,u_k}_{\mathbf{C}\text{: quadratic in }\mathbf{u}} \\ & \underbrace{-\;\gamma\Delta t\sum_{j=0}^{n-1}(T-t_j)\,u_j^\top\Sigma X_0 \;+\; \frac{\gamma(\Delta t)^3}{2}\sum_{k=0}^{n}\left(\sum_{j<k}u_j\right)^\top\Sigma\left(\sum_{j<k}u_j\right)}_{\mathbf{D_1}\text{: linear and quadratic in }\mathbf{u}} \\ & \underbrace{-\;\varrho\Delta t\sum_{j=0}^{n-1}u_j^\top\Pi X_0 \;+\; \frac{\varrho(\Delta t)^2}{2}\left(\sum_{j=0}^{n-1}u_j\right)^\top\Pi\left(\sum_{j=0}^{n-1}u_j\right)}_{\mathbf{D_2}\text{: linear and quadratic in }\mathbf{u}} \\ & +\; \underbrace{X_0^\top\mathbb{E}[P_0] + X_0^\top\mathbb{E}[A_n - A_0] - \tfrac{\gamma\Delta t}{2}(n+1)X_0^\top\Sigma X_0 - \tfrac{\varrho}{2}X_0^\top\Pi X_0}_{\text{constant, independent of }\mathbf{u}}\end{aligned} \tag{18}$$

Quadratic Form Identification. Grouping the terms above by nature and taking matrix notation, we can write $\bar{J}(\mathbf{u})$ as:

$$\bar{J}(\mathbf{u}) = -\mathbf{u}^\top \mathbf{D}^n \mathbf{u} + (\mathbf{g}^n)^\top \mathbf{u} + \text{const} \tag{19}$$

Const — from A, $\mathbf{D}_1$, $\mathbf{D}_2$.

$$\text{const} = X_0^\top \mathbb{E}[P_0] + X_0^\top \mathbb{E}[A_n - A_0] - \tfrac{\gamma \Delta t}{2}(n+1) X_0^\top \Sigma X_0 - \tfrac{\varrho}{2} X_0^\top \Pi X_0 \tag{20}$$

Linear vector $\mathbf{g}^n$ — from A, $\mathbf{D}_1$, $\mathbf{D}_2$. The coefficient of u_j from each linear term:

- **A**: u_j appears in the inner sum for all $k \geq j$, giving $\Delta t \sum_{k=j}^{n-1} \mathbb{E}[\Delta A_k] = \Delta t\, \mathbb{E}[A_n - A_j]$
- $\mathbf{D}_1$: $-\gamma \Delta t (T - t_j)\, \Sigma X_0$
- $\mathbf{D}_2$: $-\varrho \Delta t\, \Pi X_0$

Therefore:

$$g_j^n \;=\; \Delta t\, \mathbb{E}[A_n - A_j] \;-\; \gamma \Delta t (T - t_j)\, \Sigma X_0 \;-\; \varrho \Delta t\, \Pi X_0 \qquad j = 0, \ldots, n-1 \tag{21}$$

Using the telescoping identity $A_n - A_j = \sum_{k=j}^{n-1} \Delta A_k$:

$$\Delta t\, \mathbb{E}[A_n - A_j] \;=\; \Delta t \sum_{k=j}^{n-1} \mathbb{E}[\Delta A_k] \tag{22}$$

Quadratic matrix $\mathbf{D}^n$ — from B, C, $\mathbf{D}_1$, $\mathbf{D}_2$.

$$\mathbf{D}^n = \underbrace{\frac{\Delta t}{2} I_n \otimes \bar{\Lambda}}_{\mathbf{C}\text{: temporary}} + \underbrace{\frac{(\Delta t)^2}{2}(\mathbf{G}^n + (\mathbf{G}^n)^\top)}_{\mathbf{B}\text{: transient}} + \underbrace{\frac{\gamma (\Delta t)^3}{2}(L^\top L) \otimes \Sigma}_{\mathbf{D_1}\text{: risk}} + \underbrace{\frac{\varrho (\Delta t)^2}{2} J_n \otimes \Pi}_{\mathbf{D_2}\text{: terminal}} \tag{23}$$

where $(\mathbf{G}^n)_{kj} = G(t_k, t_j)\mathbf{1}_{k>j}$, $(L^\top L)_{kj} = n - max(k,j) = \frac{T - t_{max(k,j)}}{\Delta t}$

Optimal Solution. Since $\bar{J}(\mathbf{u})$ is strictly concave ($\mathbf{D}^n \succ 0$), setting the gradient to zero gives the unique maximum:

$$\boxed{\mathbf{u}^* = \frac{1}{2}\,(\mathbf{D}^n)^{-1}\,\mathbf{g}^n} \tag{24}$$

with optimal inventory $X_k^* = X_0 + \Delta t \sum_{j=0}^{k-1} u_j^*$.

3.2 Deterministic Signal Case

Let's consider price and trading signal dynamic given by:

$$dP_t = I_t\, dt + dM_t \tag{25}$$

$$dI_t = -\beta I_t\, dt, \quad \beta \in \mathbb{R}^{N \times N} \tag{26}$$

Based on that ODE, the signal is deterministic and given by:

$$I_{t_{k+1}} = e^{-\beta \Delta t} I_{t_k} \tag{27}$$

Connecting to our discrete model, $\Delta A_k = I_{t_k} \, \Delta t$. Therefore:

$$\Delta t \sum_{k=j}^{n-1} \mathbb{E}[\Delta A_k] = \Delta t \sum_{k=j}^{n-1} \mathbb{E}[I_{t_k} \Delta t] = \Delta t^2 \sum_{k=j}^{n-1} I_{t_k}$$

$$g_j^n = \Delta t^2 \sum_{k=j}^{n-1} I_{t_k} - \gamma \Delta t (T - t_j) \, \Sigma X_0 - \varrho \Delta t \, \Pi X_0 \tag{28}$$

4 Numerical Experiments

4.1 Cross-Impact Influence

Throughout this subsection the transient price impact takes the factorized form $G(t,s) = C\,\phi(t-s)\,\mathbf{1}_{\{s<t\}}$, where C is a symmetric matrix and the scalar convolution kernel ϕ is chosen among the zero kernel $\phi^{\text{zero}}(t) = 0$, the exponential kernel $\phi^{\text{exp}}(t) = e^{-\rho t}$ with $\rho = 0.5$, and the fractional kernel $\phi^{\text{frac}}(t) = t^{-\alpha}$ with $\alpha = 0.25$.

EXPERIMENTS.
Experiment 1: We consider $N = 2$ assets with initial inventory $X_0 = (10, 0)^\top$, held over a horizon $T = 10$. We impose a penalty on terminal inventory with $\varrho = 4$ and $\Pi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and we remove the risk component by setting $\gamma = 0$. We consider temporary self-impact but no temporary cross-impact through $\Lambda = \begin{pmatrix} 0.03 & 0 \\ 0 & 0.03 \end{pmatrix}$. The transient impact is generated by the three kernels above with the diagonal matrix $C = \begin{pmatrix} 0.06 & 0 \\ 0 & 0.06 \end{pmatrix}$. No trading signal is present.

Experiment 2: The setup is identical to Experiment 1, except that we now introduce cross-asset market impact through the off-diagonal entries of $C = \begin{pmatrix} 0.06 & 0.05 \\ 0.05 & 0.06 \end{pmatrix}$.

Experiment 3: The setup is as in Experiment 2, but we add a deterministic trading signal given by $dI_t = -\beta I_t\,dt$, i.e. $I_t = e^{-\beta t} I_0$, with initial signal $I_0 = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$ and decay $\beta = \begin{pmatrix} 0.9 & 0 \\ 0 & 0.3 \end{pmatrix}$. We consider both the case without cross-impact (C diagonal as in Experiment 1) and with cross-impact (C as in Experiment 2).

Experiment 4: The setup is as in Experiment 3, except that the signal decay is now $\beta = \begin{pmatrix} 0.3 & 0 \\ 0 & 0.9 \end{pmatrix}$.

Experiment 5: We extend the previous setting to $N = 3$ assets with initial inventory $X_0 = (10, 0, 0)^\top$, keeping $T = 10$, $\gamma = 0$, no trading signal, and $\varrho = 4$, $\Pi = I_3$, $\Lambda = 0.03\,I_3$. The transient impact uses a symmetric 3×3 chain cross-impact matrix

$$C = \begin{pmatrix} 0.06 & 0.05 & 0 \\ 0.05 & 0.06 & 0.05 \\ 0 & 0.05 & 0.06 \end{pmatrix},$$

so that Assets 1 and 2 interact, Assets 2 and 3 interact, while Assets 1 and 3 have no direct cross-impact.

EXPERIMENTS RESULTS.
Overall, the conclusions we draw from Experiments 1–5 are consistent with those of Abi Jaber et al. [1]. When the alpha signal is excluded, there is no significant difference between the results. When the signal is included, our plots appear as smoother versions of those presented in [1], which is expected since our signal is assumed to be deterministic. The only noticeable difference lies in the scale.

Experiment 1 Result. Without cross-impact, the two assets fully decouple. Asset 2, starting from zero inventory, is never traded: both its speed and inventory remain identically zero. Asset 1 is liquidated on its own, at roughly constant speed under the no-transient kernel, while ϕ^{exp} and

ϕ^{frac} concentrate trading near the start and the end of the window, producing almost affine inventory paths.

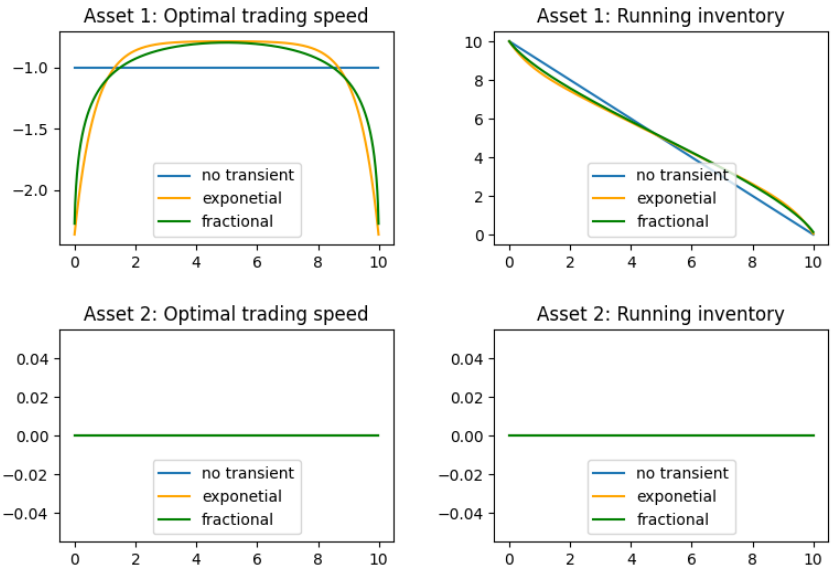


Figure 1: Impact of different kernels on the optimal trading speed and inventory of two assets in the absence of trading signal and without cross-impact

Experiment 2 Result. Switching on market cross-impact induces a *transaction-triggered round trip* in Asset 2. The mechanism is as follows: fast initial selling of Asset 1 depresses the price of Asset 2 through $C_{12} > 0$, acting as a synthetic short signal. As the selling of Asset 1 slows down, the strategy reverses and steadily buys Asset 2, since holding Asset 2 props up the price of Asset 1 via $C_{21} > 0$ and lowers the cost of its liquidation. The long position in Asset 2 is then unwound quickly near maturity. The round trip is more pronounced for ϕ^{exp} than for ϕ^{frac}. Cross-impact also renders the liquidation of Asset 1 more aggressive, so that the transient and non-transient inventory paths lie closer together than in Experiment 1.

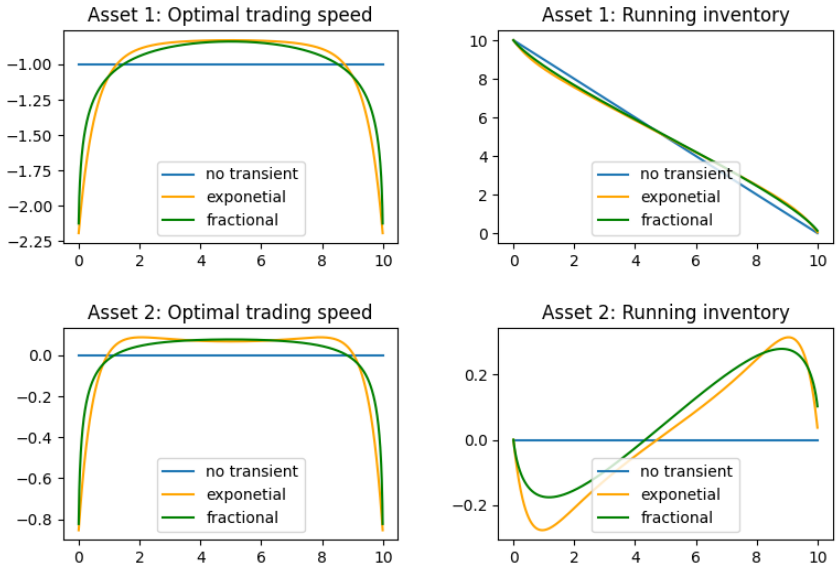


Figure 2: Impact of different kernels on the optimal trading speed and inventory of two assets in the absence of trading signal and with cross-impact

Experiment 3 Result.

- *Without cross-impact.* The positive signal on Asset 1 leads buying from the start and the positive signal on Asset 2 triggers a "buy round trip" on Asset 2 where the buying phase is more aggressive in the absence of transient impact.
- *With cross-impact.* A striking reversal occurs: despite the positive signal on Asset 1, there is almost no buying of Asset 1 at $t = 0^+$; instead the strategy sells and even shorts Asset 1 quickly. Since the signal on Asset 1 decays fast ($\beta_1 = 0.9$) while that on Asset 2 is persistent ($\beta_2 = 0.3$), shorting Asset 1 leverages cross-impact to push down the price of Asset 2, allowing the trader to accumulate more of Asset 2 cheaply and to profit from its more durable signal. As a result, more of Asset 2 is bought than in the no-cross-impact case.

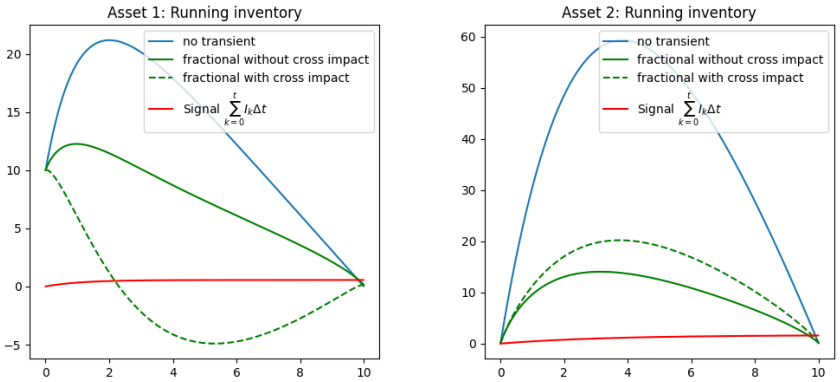


Figure 3: Impact of different kernels on the optimal trading speed and inventory of two assets in the presence of trading signal without (solid) and with (dashed) cross-impact

Experiment 4 Result. With the decay rates swapped ($\beta = \mathrm{diag}(0.3, 0.9)$), the behavior is the exact mirror of Experiment 3. Asset 1 now carries the persistent signal, so the strategy is willing to short or sacrifice Asset 2 to leverage cross-impact in favor of Asset 1. This confirms that the effect is driven by the *relative* speeds of alpha decay, and not by the labeling of the assets.

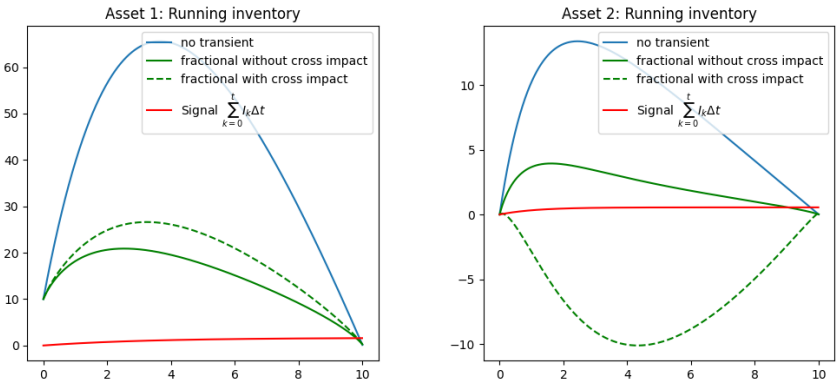


Figure 4: Impact of different kernels on the optimal trading speed and inventory of two assets in the presence of trading signal without (solid) and with (dashed) cross-impact

The general insight is that cross-impact exploits the *differential in alpha decay*: the strategy sacrifices or shorts the fast-decaying asset in order to leverage its impact on the slow-decaying one.

Experiment 5 Result. With the chain configuration coupling Assets 1 and 2 and Assets 2 and 3, but leaving Assets 1 and 3 disconnected, a *chain effect* emerges: a small-magnitude round trip in Asset 3, which has no direct interaction with Asset 1, still helps mitigate the cost of the round trip in Asset 2 through $C_{23} = C_{32}$, which in turn feeds back into the liquidation of Asset 1 through $C_{12} = C_{21}$. Cross-impact therefore propagates *transitively*: assets with no direct interaction still shape each other's optimal execution as the dimension grows.

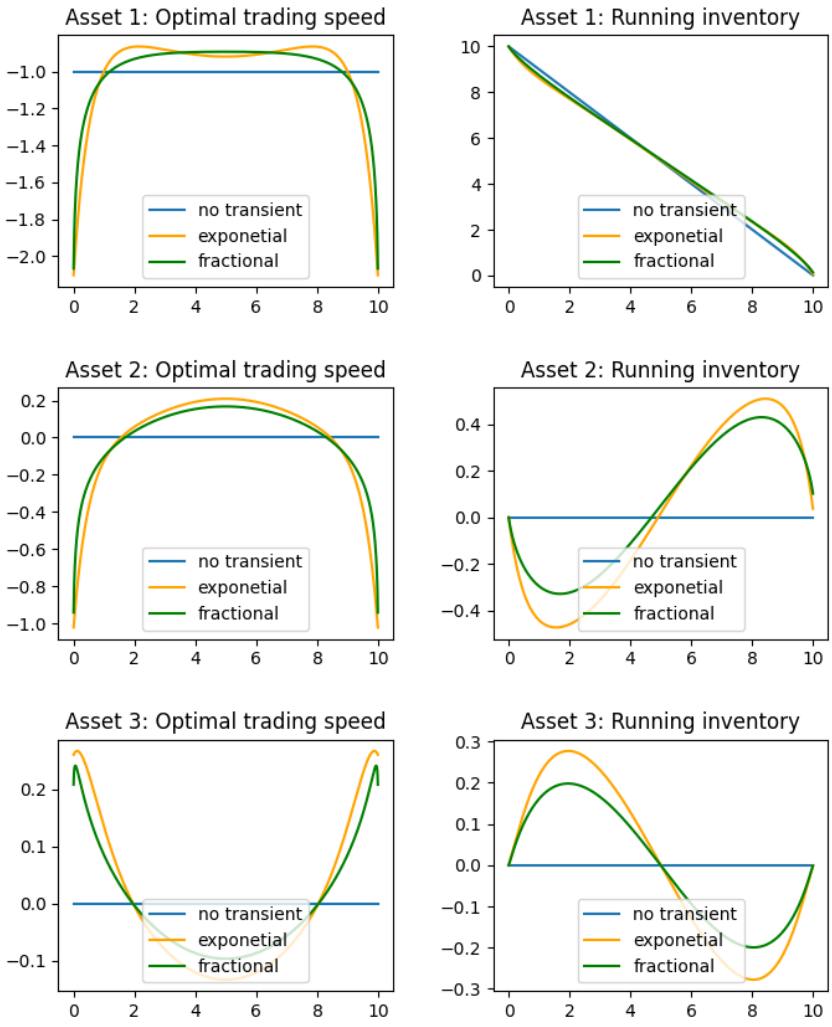


Figure 5: Impact of different kernels on the optimal trading speed and inventory of three assets in the absence of trading signal and with chain cross-impact

4.2 Influence of Correlation on Liquidation

Experiment 6: We consider the liquidation of $N = 2$ correlated assets with initial inventory $X_0 = (10, 10)^\top$ over $T = 2.5$, keeping the terminal penalty $\varrho = 4$, $\Pi = I_2$, switching off cross-impact, i.e. $C = \begin{pmatrix} 0.06 & 0 \\ 0 & 0.06 \end{pmatrix}$, and imposing no trading signal. To differentiate liquidity we set $\Lambda = \begin{pmatrix} 0.03 & 0 \\ 0 & 0.1 \end{pmatrix}$, so that Asset 1 is more liquid than Asset 2, and we introduce a risk-aversion parameter $\gamma = 0.3$. We examine two configurations of the covariance matrix: the uncorrelated case $\Sigma = 0.2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and the fully correlated case $\Sigma = 0.2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Experiment 6 Result. Figure 6 displays the inventory trajectories in both correlation config-

urations. In the uncorrelated case, the introduction of risk aversion leads to a more aggressive initial liquidation, particularly for the more liquid Asset 1. In the fully correlated case, the optimal strategy liquidates Asset 1 even more rapidly, temporarily taking a *short* position in the liquid asset in order to reduce overall portfolio risk, while unwinding the less liquid Asset 2 more gradually. This behavior is consistent with the cross-asset execution effect described in Almgren and Chriss [3] and Abi Jaber et al. in [1].

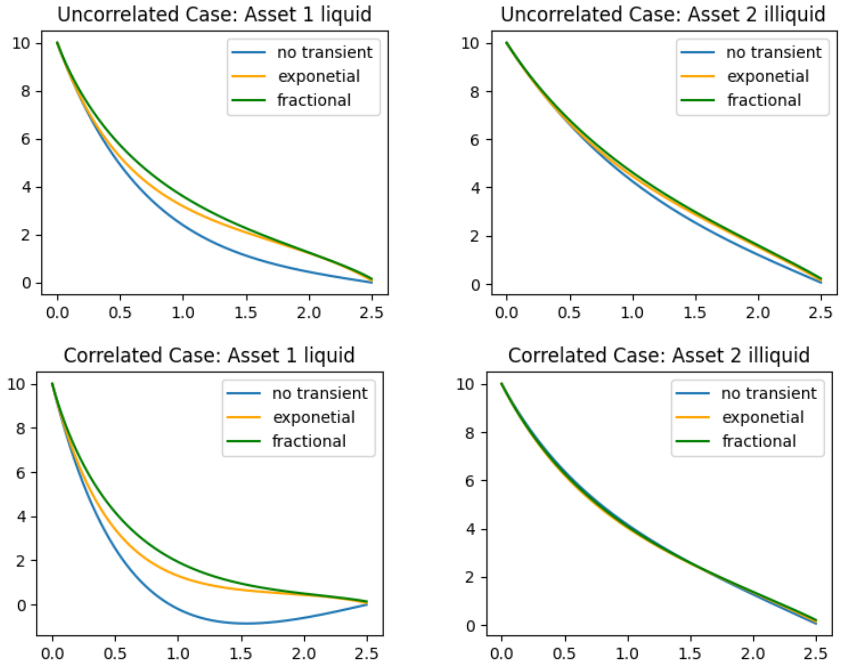


Figure 6: Impact of the correlation on the running inventory

5 Conclusion

This paper revisited the optimal portfolio selection problem of Abi Jaber et al. [1], which incorporates an alpha signal, temporary (execution) impact, and transient (market) impact with both self- and cross-impact components, and proposed an alternative, discrete-time route to its solution. By discretizing the time horizon and assuming a deterministic alpha signal, we showed that the risk-revenue functional reduces to a finite-dimensional concave quadratic program whose unique maximizer is available in closed form, $u^* = \frac{1}{2} D_n^{-1} g_n$. This avoids the stochastic Fredholm equations and resolvent operators required in the continuous-time setting, at the cost of restricting attention to deterministic signals and, consequently, to open-loop strategies.

Our numerical experiments reproduce, in discrete time, the main qualitative findings of [1]. In the absence of a signal, cross-impact alone generates transaction-triggered round trips in assets that carry no initial inventory (Experiment 2) and propagates transitively through chains of pairwise-coupled assets (Experiment 5). When signals with heterogeneous decay rates are present, the optimal strategy exploits the *differential in alpha decay*, sacrificing or shorting the fast-decaying asset to leverage cross-impact on the slower-decaying one (Experiments 3–4). Finally, asset correlation reshapes the liquidation schedule, inducing faster unwinding—and even transient short positions—in the more liquid asset in order to reduce overall portfolio risk (Experiment 6). Throughout, the strategies obtained under the deterministic-signal assumption appear as smoothed counterparts of those in [1], differing mainly in scale, which confirms the consistency of the discrete and continuous formulations.

Taken together, these results indicate that the qualitative insights of the continuous-time model carry over to the discrete-time, deterministic-signal regime, while offering a markedly simpler derivation and a transparent implementation.

Several limitations point to future work. First, the deterministic-signal assumption yields an open-loop strategy that is optimal only when the signal is deterministic; extending the framework to stochastic, adapted signals would require recovering the conditional-expectation structure of [1] and would close the gap to the fully stochastic solution. Second, a rigorous convergence analysis of the discrete scheme toward the continuous solution would strengthen the theoretical footing. Third, the optimal-trading-with-frictions experiment of Section 3.3 in [1], together with the comparison against the Markowitz benchmark, deserves to be investigated too. Finally, calibrating the propagator matrix to market data would allow an *empirical* assessment of the cross-impact effects studied here, complementing the numerical evidence presented in this paper.

References

[1] Eduardo Abi Jaber, Eyal Neuman, and Sturmius Tuschmann. Optimal portfolio choice with cross-impact propagators. 2026. Preprint, March 2026.

[2] Aurélien Alfonsi, Florian Klöck, and Alexander Schied. Multivariate transient price impact and matrix-valued positive definite functions. *Mathematics of Operations Research*, 41(3):914–934, 2016.

[3] Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3(2):5–39, 2001.

[4] Robert Almgren, Chee Thum, Emmanuel Hauptmann, and Hong Li. Equity market impact. *Risk*, 18(7):57–62, 2005.

[5] Michael Benzaquen, Iacopo Mastromatteo, Zoltán Eisler, and Jean-Philippe Bouchaud. Dissecting cross-impact on stock markets: An empirical analysis. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(2):023406, 2017.

[6] Jean-Philippe Bouchaud, Julius Bonart, Jonathan Donier, and Martin Gould. *Trades, Quotes and Prices: Financial Markets under the Microscope.* Cambridge University Press, Cambridge, 2018.

[7] Jean-Philippe Bouchaud, J. Doyne Farmer, and Fabrizio Lillo. How markets slowly digest changes in supply and demand. In Thorsten Hens and Klaus Reiner Schenk-Hoppé, editors, *Handbook of Financial Markets: Dynamics and Evolution*, pages 57–160. Elsevier, North-Holland, 2009.

[8] Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013.

[9] Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic portfolio choice with frictions. *Journal of Economic Theory*, 165:487–516, 2016.

[10] Jim Gatheral. No-dynamic-arbitrage and market impact. *Quantitative Finance*, 10(7):749–759, 2010.

[11] Victor Le Coz, Iacopo Mastromatteo, Damien Challet, and Michael Benzaquen. When is cross impact relevant? *Quantitative Finance*, 24(2):265–279, 2024.

[12] Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[13] Iacopo Mastromatteo, Michael Benzaquen, Zoltán Eisler, and Jean-Philippe Bouchaud. Trading lightly: Cross-impact and optimal portfolio execution. *Risk*, pages 82–87, July 2017.

A Appendix

A.1 Standard Markowitz Objective Function Derivation

Consider N risky assets. An investor holds a portfolio $X_t \in \mathbb{R}^N$, where X_t^i is the number of shares of asset i at time t. Let:

- $\mu_t \in \mathbb{R}^N$ be the vector of **expected returns** per share at time t,
- $\Sigma \in \mathbb{R}^{N \times N}$ be the **covariance matrix** of returns, symmetric positive definite,
- $\gamma > 0$ be the investor's **risk-aversion coefficient**.

At a static time t, the investor maximises *expected return minus a risk penalty*:

$$J(X) \;=\; \underbrace{\mu_t^\top X}_{\text{expected return}} \;-\; \underbrace{\frac{\gamma}{2} X^\top \Sigma X}_{\text{risk cost}}$$

Over a horizon $[0, T]$, expected returns and variance become stochastic, which motivates taking the expectation:

$$J(X) \;=\; \mathbb{E}\Big[\int_0^T \Big(\mu_t^\top X_t \;-\; \frac{\gamma}{2} X_t^\top \Sigma X_t\Big)\; dt\Big]$$

Integration by parts. Let $u_t = \dot{X}_t$ denote the **trading speed**, so that $dX_t = u_t\, dt$. The unaffected price decomposes as $P_t = A_t + M_t$ where:

- $dA_t = \mu_t\, dt$ (the predictable drift / signal),
- M_t is a continuous martingale with $d[M^i, M^j]_t = \Sigma_{ij}\, dt$.

Applying the product rule to $X_t^\top P_t$:

$$d(X_t^\top P_t) \;=\; dX_t^\top P_t + X_t^\top\, dP_t \;=\; u_t^\top P_t\, dt + X_t^\top\, dP_t$$

Integrating from 0 to T:

$$X_T^\top P_T \;-\; X_0^\top P_0 \;=\; \int_0^T u_t^\top P_t\, dt + \int_0^T X_t^\top\, dP_t$$

Rearranging gives:

$$-\int_0^T u_t^\top P_t\, dt + X_T^\top P_T \;=\; X_0^\top P_0 + \int_0^T X_t^\top\, dP_t$$

Taking expectations and using the fact that stochastic integrals against martingales have zero expectation:

$$\mathbb{E}\Big[\int_0^T X_t^\top\, dP_t\Big] \;=\; \mathbb{E}\Big[\int_0^T X_t^\top \mu_t\, dt\Big] + \underbrace{\mathbb{E}\Big[\int_0^T X_t^\top\, dM_t\Big]}_{=\,0}$$

Therefore:

$$\mathbb{E}\Big[\int_0^T X_t^\top \mu_t\, dt\Big] \;=\; \mathbb{E}\Big[-\int_0^T u_t^\top P_t\, dt + X_T^\top P_T\Big] - X_0^\top P_0$$

As $X_0^\top P_0$ is a constant (does not depend on the strategy), maximising $\mathbb{E}\Big[\int_0^T \big(\mu_t^\top X_t - \frac{\gamma}{2} X_t^\top \Sigma X_t\big)\; dt\Big]$ is equivalent to maximising:

$$\mathbb{E}\Big[-\int_0^T u_t^\top P_t\, dt + X_T^\top P_T - \frac{\gamma}{2}\int_0^T X_t^\top \Sigma X_t\, dt\Big]$$

A.2 Paper Objective Function Derivation

Adding temporary impact and transient impact modifies the effective execution price at which each trade is executed:

$$P_t \;\longrightarrow\; P_t + D_t^u + \tfrac{1}{2}\Lambda u_t$$

where $D_t^u = \int_0^t G(t,s)u_s\, ds$ is the accumulated transient price distortion and $\frac{1}{2}\Lambda u_t$ is the instantaneous temporary impact. Substituting into the objective above yields the full objective functional of the paper:

$$J(u) \;=\; \mathbb{E}\Big[-\int_0^T u_t^\top \big(P_t + D_t^u + \tfrac{1}{2}\Lambda u_t\big)\, dt + X_T^\top P_T - \frac{\gamma}{2}\int_0^T X_t^\top \Sigma X_t\, dt\Big]$$

A.3 Quadratic Optimisation — General Case

Let $A \in \mathbb{R}^{N\times N}$, $b \in \mathbb{R}^N$, and $c \in \mathbb{R}$. Consider the **unconstrained quadratic maximisation**:

$$\max_{X\in\mathbb{R}^N} \quad f(X) \;:=\; -X^\top AX + b^\top X + c$$

The gradient and Hessian of f are:

$$\begin{aligned}\nabla_X f &= -(A + A^\top)X + b\\ \nabla_X^2 f &= -(A + A^\top)\end{aligned}$$

Requirement for existence of the solution.
For a **unique global maximum** to exist, f must be **strictly concave**, which requires the Hessian to be negative definite:

$$\nabla_X^2 f \;\prec\; 0 \iff A + A^\top \succ 0$$

- $A + A^\top \succ 0$ is a **condition that must be verified** for any given A — it does not hold automatically. For instance, $A = -I$ gives $A + A^\top = -2I \prec 0$, so no maximum exists.
- When A is **symmetric** ($A = A^\top$), the condition simplifies to $A \succ 0$ (strictly positive definite).
- b can be any vector in $\mathbb{R}^N$ once the condition on A is satisfied.

Derivation of the Solution.
Setting $\nabla_X f = 0$ implies:

$$X^* \;=\; \begin{cases}(A + A^\top)^{-1}b & \text{if } A \text{ is any matrix with } A + A^\top \succ 0\\ \dfrac{1}{2}\,A^{-1}b & \text{if } A \text{ is symmetric with } A \succ 0\end{cases}$$

A.4 Markowitz Optimisation Solving

Static case. The fixed-time Markowitz problem is exactly the quadratic problem above with the identification:

$$A \;=\; \frac{\gamma}{2}\,\Sigma, \qquad b \;=\; \mu_t, \qquad c \;=\; 0$$

Since Σ is symmetric positive definite and $\gamma > 0$, the matrix $A = \frac{\gamma}{2}\Sigma$ is **symmetric positive definite**. The symmetric case therefore applies:

$$X^* \;=\; \frac{1}{2}A^{-1}b \;=\; \frac{1}{2}\cdot\frac{2}{\gamma}\,\Sigma^{-1}\mu_t \;=\; \frac{1}{\gamma}\,\Sigma^{-1}\mu_t$$

Dynamic discrete case. Fix $n \in \mathbb{N}$ and partition $[0, T]$ into $n+1$ equally spaced dates:

$$t_0 = 0 \;<\; t_1 \;<\; \cdots \;<\; t_n = T, \qquad \Delta t \;:=\; \frac{T}{n}$$

The discrete frictionless objective is:

$$J(X) \;=\; \sum_{k=0}^{n} \left(\mu_k^\top X_k \;-\; \frac{\gamma}{2} X_k^\top \Sigma X_k \right) \Delta t \tag{29}$$

Since the objective is separable across time steps — there are no frictions, so X_k and X_j for $k \neq j$ are not coupled — the problem decomposes into $n+1$ independent static problems, one at each t_k. Applying the static solution at each step:

$$X_k^* \;=\; \frac{1}{\gamma} \Sigma^{-1} \mu_k, \qquad k = 0, 1, \ldots, n \tag{30}$$

Continuous time limit. As $n \to \infty$ ($\Delta t \to 0$), the discrete solution converges to:

$$X_t^* \;=\; \frac{1}{\gamma} \Sigma^{-1} \mu_t, \qquad t \in [0, T] \tag{31}$$

Optimal trading speed. If we denote $\dot{X}_t = \frac{dX_t}{dt}$, then $\dot{X}_t = u_t$ and the optimal trading speed in the frictionless case is:

$$u_t^* \;=\; \frac{1}{\gamma} \Sigma^{-1} \dot{\mu}_t$$

This formula is valid when μ_t is deterministic and differentiable. When μ_t is stochastic, $\dot{\mu}_t$ does not exist and one must write $du_t^* = \frac{1}{\gamma} \Sigma^{-1} \, d\mu_t$ in the Itô sense. In both cases, u_t^* is potentially unbounded, motivating the addition of temporary impact Λ and transient impact G which regularise the optimal speed into a finite, well-defined strategy.